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REVIEW

Large language models in the vertical integration of spine surgery workflow: a scoping review

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Abstract

Purpose To evaluate current evidence regarding the clinical reliability and reasoning capabilities of Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs) within the spine surgery workflow. This scoping review utilizes a novel Vertical Integration Maturity Scale (VIMS) to map model maturity across five stages of the vertical workflow, identifying persistent research gaps and technical prerequisites for clinical implementation.

Methods A systematic search was conducted across PubMed, Embase, and Scopus for peer-reviewed studies published between January 2023 and January 2026. Utilizing the Population, Concept, and Context (PCC) framework, studies were selected based on their application of generative AI to the clinical evaluation and surgical management of spinal pathologies. Data were analyzed via independent dual-coding using a two-dimensional framework mapping five VIMS levels across five functional workflow stages. Synthesis included categorization by model architecture, input modality, and performance benchmarks, with inter-rater reliability calculated to validate the novel framework.

Results Of 351 identified records, 40 studies met the inclusion criteria. Research density peaked in Stage III (Evaluative Decision Logic) at VIMS Level 1 ($n=15$), indicating a methodological focus on evidence-based guideline retrieval over patient-specific synthesis. While ChatGPT-4 showed high concordance (61.1% to 88.2%) with clinical guidelines, diagnostic precision fell to 11.1% in complex spinal deformity. A critical evidence gap persists in Stage IV (Tactile Procedure Planning) with a total absence of studies evaluating autonomous workflow agency (VIMS 4–5). This absence likely reflects a limitation of retrospective study designs reliant on proprietary, non-version-controlled models. While qualitative synthesis suggests hybrid architectures and Retrieval-Augmented Generation (RAG) may bridge existing semantic gaps, the inferential strength of these trends is currently limited by heterogenous benchmarking metrics.

Conclusion Current LLMs demonstrate proficiency in foundational medical knowledge, but a significant evidence gap remains regarding their inferential validity in the non-deterministic surgical scenarios. Advancing clinical integration requires modular multimodal ensembles that process radiographic data alongside domain-specific reasoning scaffolds. Crucially, future research must transition from retrospective algorithmic benchmarking toward standardized, prospective clinical trials to evaluate algorithmic decision-making against real-world surgical outcomes.

Keywords Spine surgery · Large language models · Artificial intelligence · Clinical decision-support systems · Surgical workflow · Vertical integration

Introduction

The rapid advancement of Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs) has prompted investigation into their integration within specialized surgical workflows, offering transformer-based architectures capable of synthesizing heterogenous medical datasets for advanced decision support [1–3].

Spine surgery represents a primary domain for Artificial Intelligence (AI) integration due to the increasing prevalence

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of degenerative pathologies and the inherent complexity of surgical decision-making [4, 5]. Although evidence-based guidelines offer a structured approach, the interpretation of patient-reported outcomes and multimodal datasets is often limited by significant inter-observer variability [6].

High-dimensional computational systems may address this by synthesizing complex variables, such as diagnostic imaging and gait analysis, to facilitate patient-specific planning that complements traditional surgical judgment. Within this framework, LLMs offer the potential for reproducible data synthesis that may enhance clinical consistency [7]. By augmenting human decision-making, these technologies facilitate a personalized paradigm where objective analysis supports clinical expertise.

A growing number of medical professionals are seeking to incorporate LLMs in clinical practice. However, the reliability and inferential validity of these models remain largely unvalidated. While existing literature describes general AI trends, a critical research gap persists regarding the mapping of evidence concerning the reasoning capabilities of LLMs within complex, multi-step environments [8].

This scoping review utilizes a novel Vertical Integration Maturity Scale (VIMS) framework to map the current evidence across five distinct stages of the surgical workflow, from initial triage to perioperative management. We define vertical integration as the implementation of clinical decision-support systems capable of navigating multi-step workflows under human supervision. By utilizing this dual-track framework, we clarify how current models align with VIMS levels, ranging from foundational recall to autonomous partnership.

Guided by the Population, Concept, and Context (PCC) framework, this study evaluates the capacity of LLMs and MLLMs to process history, examination, and imaging data to formulate diagnostic and therapeutic plans for spinal pathologies.

Table 1 Vertical Integration Maturity Scale (VIMS)

Level	Maturity	Definition
VIMS 1	Foundational Recall	Retrieval of evidence-based guidelines and medical knowledge without patient-specific synthesis.
VIMS 2	Pattern Recognition	Categorization of discrete clinical features and radiographic findings from raw or structured data.
VIMS 3	Evaluative Synthesis	Integration of multimodal variables to generate definitive clinical judgments and personalized management strategies.
VIMS 4	Workflow Agency	Execution of multi-step clinical tasks and actionable plan within complex environments.
VIMS 5	Autonomous Partnership	Closed-loop collaboration with human specialists to optimize intraoperative and longitudinal outcomes.

Methods

The study was conducted in accordance with PRISMA-ScR guidelines [9]. This methodology was selected to systematically identify persistent research gaps and limitations in the clinical reliability of LLMs within the spine surgery workflow.

Search strategy

A systematic search was executed across PubMed, Embase, and Scopus. The search string utilized Boolean operators to integrate keywords and MeSH pertaining to Large Language Model and spine surgery (Supplementary Material 3).

Eligibility criteria

Study eligibility was defined utilizing the PCC framework. The population included patients undergoing evaluation or surgery for spinal pathologies. The concept was LLM and MLLM application. The context was vertical clinical workflow implementation from initial triage to postoperative recovery. Inclusion was restricted to peer-reviewed studies published in English between January 2023 and 21 January 2026. Studies lacking an LLM component or focusing exclusively on non-generative AI architectures were excluded.

Study selection

Two blinded reviewers independently performed screening and full-text eligibility evaluations based on the PCC framework, resolving discrepancies through consensus, as documented in the PRISMA flow diagram.

Data extraction and synthesis

Data were extracted using a standardized table to capture study characteristics, model architecture, assigned VIMS levels, workflow stage classifications, performance metrics and limitations (Supplementary Material 1). To address the absence of standardized metrics for evaluating AI agency in specialised surgical environments, we developed and applied a novel two-dimensional analytical framework designed to bridge the gap between technical model performance and practical surgical utility.

The first dimension utilized the VIMS to quantify the depth of clinical reasoning. These five levels range from foundational recall to autonomous partnership with human specialists (Table 1). The second dimension mapped these maturity levels across five discrete stages of the vertical surgical workflow, from initial triage to perioperative management

Table 2 Stages of the Vertical Surgical Workflow

Stage	Domain	Definition
Stage I	Initial Triage	Aggregation of history and physical examination findings for clinical triage.
Stage II	Multimodal Correlation	Alignment of patient symptoms with objective diagnostic evidence and radiographic findings.
Stage III	Evaluative Decision Logic	Synthesis of clinical variables to determine treatment plans and surgical candidacy.
Stage IV	Tactical Procedure Planning	Formulation of a specific surgical constructs, including approach selection, instrumentation, and alignment targets.
Stage V	Perioperative Management	Complication risk stratification and longitudinal monitoring of patient recovery.

(Table 2). This dual-axis approach facilitates a granular assessment of model reliability and identifies research gaps and current limitations within the LLM-assisted spine surgery workflow.

Because included studies may span across multiple Vertical Surgical Workflow stages, two authors independently coded all applicable stages for each study. For Cohen κ statistical reliability analysis, which requires mutually exclusive categories, raters assigned a single primary stage based strictly on the primary methodological hypothesis of each study.

Results

Study selection and characteristics

The systematic search yielded a total of 351 records, of which 40 studies met the final inclusion criteria (Fig. 1). A

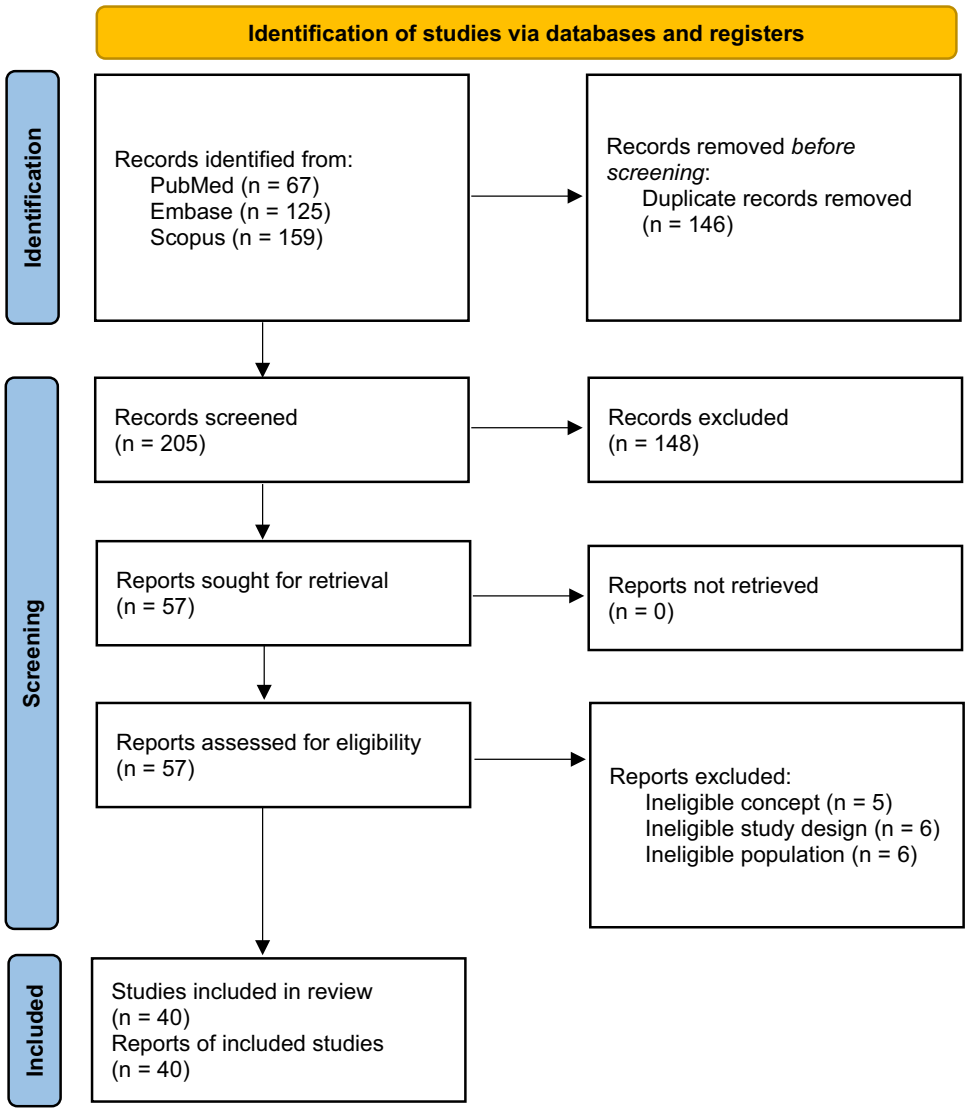


Fig. 1 PRISMA flow diagram

	VIMS 1	VIMS 2	VIMS 3	VIMS 4	VIMS 5
Stage I	2	1	3	0	0
Stage II	2	8	1	0	0
Stage III	15	5	6	0	0
Stage IV	0	1	1	0	0
Stage V	3	0	2	0	0

Fig. 2 Distribution of LLM research across workflow stages and VIMS levels with colour intensity representing study frequency. Note: To capture the full breadth of the assesses areas, studies encompassing multiple workflow domains were counted across all applicable stages based on the consensus coding

comprehensive summary of the extracted data is provided in the Supplementary Table.

The literature demonstrates a significant acceleration in research activity, progressing from a single publication in 2023 to 14 in 2024 and peaking at 24 in 2025. Evaluations focused predominantly on proprietary OpenAI GPT series (n = 31), including GPT-4 (n = 21), GPT-3.5 (n = 13), and GPT-4o (n = 11). Other evaluated models included Google Gemini (n = 7) and Anthropic Claude (n = 2). While foundational models were the primary focus, a few studies utilized Retrieval-Augmented Generation (RAG) to ground model outputs in evidence-based spinal surgical guidelines, and a minority of studies integrated Convolutional Neural Network (CNN) architectures to facilitate multimodal correlation [10–14]. Most studies evaluating proprietary systems utilized broad model identifiers without documenting specific version-locked API endpoints, introducing longitudinal variability into the aggregate dataset.

Independent dual coding of the 40 included studies demonstrated high inter-rater reliability. Utilizing the primary stage assignments, agreement for the Vertical Surgical Workflow classification was $\kappa = 0.74$, and agreement for the VIMS assignment was $\kappa = 0.88$. All classification discrepancies were resolved through consensus discussion (Supplementary Material 4).

Mapping the included studies across the analytical framework reveals marked clustering within lower maturity levels across early and mid-stage workflow (Fig. 2). Research density is highest in Stage III at VIMS 1 ($n = 15$), indicating that current study designs focus predominantly on the static retrieval of evidence-based guidelines rather than testing patient-specific inferential synthesis.

A significant evidence gap exists in Stage IV, which was addressed by only 2 studies at VIMS 2. This paucity likely reflects the technical complexity of formulating specific surgical constructs and the prerequisite for sophisticated multimodal imaging interpretation in study design. No studies evaluated applications at VIMS 4 or VIMS 5 across any workflow stage.

Analysis by vertical workflow stage

Stage I: Initial triage

Six studies evaluated initial triage capabilities utilizing standardized clinical queries or structured vignettes. Direct interpretation of raw radiographic imaging remained unaddressed at this stage. ChatGPT-4 and GPT-4o demonstrated diagnostic accuracy reaching 100% within high-quality text-based vignettes [15]. Notably, LLM-generated management plans exceeded those of board-certified surgeons in comprehensiveness (+ 34%; $p = 0.0176$) [16]. However, superior structural formatting and qualitative presentation scores did not correlate with clinical precision or adherence to evidence-based guidelines [16, 17].

In cauda equina syndrome triage, concordance with multidisciplinary teams reached 99.7% ($\kappa = 0.764$) [11]. Despite this, a systemic trend toward over-referral was noted across Stage I studies. This was characterized by high sensitivity for surgical urgency but suboptimal specificity, potentially increasing specialised referral burdens [11, 18]. Studies indicated that iterative model updates have largely mitigated the source fabrication and hallucinations associated with ChatGPT-3.5. Subtle inconsistencies with established clinical protocols persist [17, 19], and diagnostic reliability remains secondary to structural consistency and formatting [16].

Stage II: Multimodal correlation

Eleven studies evaluated the correlation of clinical symptoms with multimodal data, such as MRI and CT imaging. Evidence demonstrates a transition toward multimodal feature extraction, however, unified diagnostic synthesis

integrating raw imaging with clinical findings remains unachieved [20, 21].

Standalone LLMs exhibited significant limitations in visuospatial tasks. In landmarking for adolescent idiopathic scoliosis, agreement with experts was negligible ($\kappa \leq 0.053$), with > 73% of anatomical assignments identified as structurally incorrect [21]. Models failed to reliably localize vertebral levels or quantify pathological severity ($\kappa = 0.364$) without external software integration [20, 22].

Within individual evaluations, specialized architectures achieved higher sensitivity in narrow diagnostic tasks, whereas standalone generalist models demonstrated suboptimal performance (~ 70%) for identical applications [13]. Conversely, specialized vision-language models demonstrated high internal consistency but failed in general recognition tasks due to narrow training parameters [14].

Emerging evaluations of hybrid systems indicate that integrating specialized ML with LLM inferential reasoning yields strong clinical concordance in stenosis classification ($\kappa = 0.849$) [23] and high predictive accuracy (F1 = 0.905) for complex disorder combinations [12]. However, owing to the heterogeneity of reported performance metrics across the literature, these architectural comparisons represent descriptive trends rather than statistical certainties. Crucially, despite these performance gains, 30% of high-acuity recommendations within complex diagnostic pathways remained clinically unsafe [24].

Stage III: Evaluative decision logic

Twenty-two studies evaluated the concordance of evidence-based guidelines and integrating patient-specific factors to determine surgical candidacy.

In sixteen studies utilizing standardized guideline-based prompts, assessing VIMS 1, ChatGPT-4 and ChatGPT-4o achieved concordance rates between 61.1% and 88.2% [25–29]. While specific lumbar fusion models exceeded board-certified benchmarks (88.2% vs. 72.6%) [26], performance remained inconsistent. Other studies reported lower concordance (42–59%) primarily due to incomplete responses and omission of critical guideline factors [30–32].

Nine studies utilized clinical vignettes, assessing VIMS 3 level, revealing a maturity gap between theoretical knowledge and expert-level judgement (3.90 vs. 4.70 Likert score) [33]. Diagnostic precision was inversely correlated with case complexity, falling to 11.1% in complex spinal deformity [34, 35]. Despite 73% alignment with multidisciplinary teams in spinal oncology, inter-rater reliability for surgical intervention remained below human consensus ($\kappa = 0.548$) [36].

Stage IV: Tactile procedure planning

Two studies evaluated model capacity for surgical construct selection and biomechanical planning. Kartal et al., [37] identified variance in procedural recommendations, characterized by a failure to integrate complex variables such as spinal stability and prior surgical history which resulted in clinically inappropriate selections. In adolescent idiopathic scoliosis, model-expert agreement was negligible ($\kappa \leq 0.053$ vs. $\kappa = 0.913$) [21]. Precision was fundamentally compromised by structurally invalid anatomical landmarking (73.6%-80.8% error rate), rendering automated Cobb angle quantifications and subsequent procedural planning clinically invalid [21].

Stage V: Perioperative management

Five studies evaluated postoperative risk stratification and complication management. Successive model iterations demonstrated performance gains. Concordance with NASS guidelines for thromboembolic and antibiotic prophylaxis improved from 33 to 62.5% in ChatGPT-3.5 to 81–92% in ChatGPT-4.0 [38, 39]. In pediatric surgical site infection (SSI) prevention, concordance with expert consensus reached 92.8% [40].

Current evidence indicates a transition toward predictive modelling utilizing unstructured datasets. The specialized GatorTron LLM architecture outperformed traditional computer vision models in forecasting sepsis and pulmonary events, achieving a specificity of 0.946 for neurological complications [41]. For spinal SSI management, ChatGPT-4 synthesized heterogenous data, including history, radiographic descriptions, biochemical markers, and microbiology with high expert-rated precision ($\kappa = 0.62–0.83$) [42].

Discussion

Analysis of the 40 included studies demonstrates a transition from passive information retrieval toward active clinical synthesis. This progression signals the repositioning of LLMs from simple query tools to potentially functional components of the five-stage spinal surgical workflow. However, clinical integration is currently constrained by a series of interconnected maturity gaps. Although LLMs demonstrate high guideline adherence, current evaluations reveal a deficit in the clinical contextualization required for complex pathologies, resulting in diagnostic ambiguity or unwarranted clinical certainty. Furthermore, most investigations remain limited by unimodal text-only evaluations that cannot process multimodal morphometric data effectively, precluding true surgical planning. Technical instabilities,

specifically prompt-dependency and hallucinations further necessitate continuous human oversight to ensure safety. Addressing these architectural and clinical barriers is essential for vertical workflow integration.

The clinical reasoning gap: Guidelines, heuristics, and decisiveness

A primary conflict identified in this review is the discrepancy between LLM outputs and established clinical practice guidelines. While 19 studies utilize guideline concordance as a benchmark, this approach assumes guidelines represent a static gold standard, often ignoring the developmental latency between primary research and formal codification [26, 43]. Interestingly, while models often demonstrate higher concordance rates than board-certified surgeons, they lack the nuanced flexibility required for complex surgical decision-making [26]. Surgeons frequently deviate from rigid protocols based on patient-specific factors, with compliance for procedures such as lumbar fusion reaching only 60% in clinical practice [6]. Since surgical practice remains heterogenous [6, 26, 44] but adherence to evidence-based pathways correlates with superior outcomes [45], we believe that vertical integration depends not on the automation of static guidelines, but on the creation of verified medical knowledge bases for dynamic, real-time querying [31].

The transition from VIMS Level 1 to Level 3 is further obstructed by reasoning decay, where LLM reliability collapses during the multi-step integration of complex patient narratives. This decay occurs precisely when unstructured data, such as symptom chronicity and psychosocial stability, must be synthesized into a personalized surgical candidacy. Standard architectures frequently suffer from error propagation, subtle misinterpretations during initial data aggregation led to significant management failures. Furthermore, a distinct heuristic barrier exists where models function as efficient data retrievers but fail to replicate fellowship-level clinical gestalt [33, 34, 38, 39]. Expert surgeons utilize heuristics derived from longitudinal experience to weigh variables and act preemptively [11, 26]. Current LLMs tend to treat these critical modifiers as clinical noise, retreating into generic disclaimers or rigid guideline adherence [11, 27]. Future research should prioritize iterative prompting strategies with Chain-of-Thought (CoT) scaffolds designed to sustain long-form clinical logic, rather than relying on the unidirectional data inputs of traditional chatbot interfaces [19]. Ultimately, the capacity for nuanced clinical contextualization remains the prerequisite for the safe integration of AI into spinal practice.

Results reveal a decisiveness paradox, while fellowship-trained surgeons demonstrate clinical decisiveness in over 95% of complex scenarios, LLMs remain decisive in as few as 24% [25, 36, 38]. Rather than acting as assertive clinical partners, models tend toward algorithmic conservatism when faced with non-standardized data [36, 46]. This hesitancy prevents current LLMs from functioning as reliable adjuncts in the controversial or emergency settings that define high-stakes spine surgery [15].

Conversely, LLMs occasionally exhibit artificial over certainty, fabricating recommendations to satisfy a prompt, a failure characterized by an inability to admit knowledge limits [25, 27, 29, 30, 40, 47, 48]. This confabulated certainty presents a significant risk to patient safety. In the absence of direct textual precedents, models may lack the cognitive agency required to propose targeted surgical interventions [38, 47]. To achieve VIMS 3 or 4, systems must incorporate confidence-calibration mechanisms that automatically flag ambiguous scenarios for human escalation. Bridging this gap requires a move toward models capable of out-of-distribution inference, grounded in domain-specific fine-tuning rather than probabilistic adherence to explicit training parameters.

Evidentiary integrity and structural reliability

The clinical implementation of LLMs is fundamentally obstructed by a disconnect between probabilistic reasoning and a validated evidence base. This risk is primarily driven by data contamination. Training on heterogenous datasets causes LLMs to oscillate between outdated protocols and recovery bias, prioritizing unproven trials over established safety benchmarks [24, 27–29, 49, 50]. Currently, LLMs lack the capacity of autonomous critical appraisal, frequently prioritizing high-volume data over methodological rigor [18, 24]. In rapidly evolving domains such as minimally invasive and endoscopic spine surgery, this inability to weight evidence remains a definitive barrier to VIMS Level 3 integration.

This evidentiary gap is exacerbated by artificial hallucinations, specifically citation fabrication. Despite iterative reductions in error frequency, this residual risk remains a critical liability in spine surgery [7, 25, 48]. Consequently, current architectures cannot function as autonomous authorities and necessitate a mandatory human-in-the-loop protocol. Rigorous specialist verification of LLM-generated clinical plans is a prerequisite for ensuring data accuracy and navigating the complex medicolegal requirements of modern spinal practice.

The reliability of these outputs is structurally compromised by heterogenous prompt designs. Highly structured prompts often induce prompt overfitting by utilizing pre-processed vignettes that isolate diagnostic features while omitting the data noise inherent in real-world practice [15, 35, 47]. This raises significant concerns regarding

ecological validity, as current benchmarks frequently utilize pre-filtered inputs that inadvertently signal the very heuristics being tested, failing to replicate the independent variable-weighting required of human specialists [35, 51–53].

Reproducibility challenges in proprietary models exacerbate this structural fragility. Since most systems undergo continuous, undisclosed algorithmic updates, the evidence base is vulnerable to temporal degradation. Utilizing broad model identifiers without version control means studies published months apart may evaluate fundamentally different algorithms, precluding reliable cross-study validation. As the field advances, future clinical benchmarks must prioritize open-source architectures or mandate the use of strictly versioned API endpoints to ensure longitudinal scientific integrity.

Finally, system reliability is compromised by a documented self-correction deficit. Once an initial logical failure occurs, models exhibit a statistical bias toward their own erroneous outputs, rendering iterative dialogue an unreliable method for error rectification [17]. Consequently, clinical safety cannot rely on autonomous self-correction, making the precision of the initial prompt the critical determinant of clinical safety. Addressing these inconsistencies requires a transition toward standardized surgical reasoning benchmarks that simulate the diagnostic complexity and noise of the spine clinic. This methodological shift must move beyond the unidirectional, clean data inputs that currently dominate the literature and critically evaluate whether the current chatbot interface is suitable for integration into the vertical workflow [17, 54].

Bridging the semantic gap: Hybrid architectures and RAG

The current trajectory of vertical integration exhibits an observed performance ceiling. While this limitation is partly dictated by the unimodality of text-based architectures, it is also heavily influenced by a sampling artifact within the existing literature, which predominantly features retrospective, lower-maturity evaluations. Although eleven studies indicate a nascent transition toward multimodal feature extraction; however, a profound semantic gap remains, wherein high linguistic proficiency frequently obscures a fundamental deficit in anatomical grounding and spatial reasoning [7, 20–22]. To achieve VIMS Level 4, the field must transition from monolithic designs toward modular hybrid ensembles that decouple spatial-geometric computation from contextual reasoning [21], validated through prospective clinical study designs.

Within these emerging frameworks, specialized ML models, such as Convolutional Neural Networks (CNNs) optimized for image segmentation, can execute high-precision spatial tasks, allowing the LLM to function as a central orchestrator that synthesizes these findings into patient-specific plans [13]. Although heterogenous reporting metrics preclude definitive cross-study statistical meta-analysis, qualitative synthesis suggests that integrating morphological parameters as discrete tokens within a multimodal ensemble provides a viable pathway to overcome the reasoning gap that currently restricts standalone agents to VIMS Level 3 [12, 23].

Furthermore, RAG represents a promising architectural shift to address the evidence integrity gap. By anchoring reasoning in curated, real-time databases, RAG enables models to prioritize contemporary primary literature over the legacy snapshots inherent in static training data, thereby reducing the frequency of artificial hallucinations [10, 24, 37, 50]. Although the inferential strength of this claim is currently limited by the lack of commensurable benchmarks across studies, this data synchronization facilitates the transition from algorithmic opacity toward a verifiable engine for clinical decision support [25, 50].

Ethical governance and accountability

While a comprehensive ethical analysis is beyond the scope of this review, the legal and ethical implications of integrating AI into surgical workflows are paramount. The clinical integration of LLMs into spinal surgery is fundamentally constrained by algorithmic opacity, which obstructs the transparent justification of recommendations required for surgical accountability [48, 55]. Beyond reasoning transparency, these models introduce significant liabilities regarding data sovereignty, legal responsibility, and the potential for perpetuating systemic biases inherent in uncurated training datasets [56, 57]. Consequently, maintaining a safe surgical workflow remains contingent upon mandatory human-in-the-loop oversight to ensure model outputs synchronize with evolving medicolegal requirements and standards of care [58].

Limitations

This scoping review is primarily limited by a focus on English-language literature and general-purpose LLMs, potentially omitting proprietary or specialized architectures utilized in private institutional settings. There is a significant paucity of evidence regarding domain-specific models fine-tuned on biomedical corpora. Furthermore, the rapidly evolving landscape of generative AI suggests that the most recent iterations may exhibit capabilities not fully captured in the current dataset.

Crucially, all 40 included studies rely entirely on retrospective benchmarking, clinical vignettes, or simulated

datasets. The complete absence of prospective patient cohorts or randomized controlled trials fundamentally limits the ability to evaluate algorithmic decision-making against real-world clinical outcomes. Additionally, the reliance on heterogenous benchmarking (e.g. concordance percentages, Cohen κ, F1 scores, Likert scales) precludes a formal quantitative meta-analysis. Consequently, conclusions regarding the relative superiority of specific architectural designs, such as hybrid ensembles or RAG, are drawn from metrically non-commensurable cross-study comparisons. This metric heterogeneity weakens the inferential strength of these comparative claims, necessitating a cautious interpretation of current performance trends until standardized, domain-specific evaluations frameworks are established. To address this methodological gap, our novel framework proposed herein represents a necessary first step toward building a standardized evaluative methodology. It is designed to unify future research reporting and safely accelerate the evaluation of clinically integrated AI across distinct studies and research centers. Ultimately, continuous collaborative efforts are required to further standardize research methodologies within this domain.

A fundamental challenge in applying the proposed framework was capturing the comprehensive breadth of the investigated literature. We intentionally utilized a multistage classification approach for visual synthesis, as restricting comprehensive studies to a single domain artificially underrepresents their clinical coverage. Although a primary stage was assigned for the purpose of validating inter-rater reliability, readers should recognize that this classification framework is designed to map the aggregate landscape of surgical artificial intelligence. It is not intended to strictly isolate studies or limit their interpretation to the narrow scope of a singular methodological hypothesis.

Conclusion

This synthesis identifies a critical paradigm shift in spinal AI, transitioning from passive information retrieval toward autonomous clinical synthesis. While frontier LLMs demonstrate high proficiency in foundational medical knowledge, the current literature reveals a significant evidence gap regarding their inferential validity in the non-deterministic scenarios of surgical practice. Achieving true workflow agency and autonomous partnership requires the transitioning beyond unimodal linguistic frameworks.

Future integration depends on the development of hybrid multimodal ensembles that natively integrate DICOM data and longitudinal patient metrics with specialized hierarchical reasoning scaffolds. Establishing a validated, prospective evidence base is necessary for safe implementation. Crucially, future research priorities must shift from retrospective algorithmic benchmarking toward rigorous clinical validation. Establishing prospective cohorts and randomized controlled trials to evaluate algorithmic decision-making against actual surgical outcomes is an absolute prerequisite for safe clinical implementation.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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